A Practical Guide to Gaussian process

TOPIC -- GAUSSIAN PROCESS

$$f(x) \sim \mathcal{GP}(m(x), k(x, x'))$$

-- 01 --

Stationary case For a stationary Gaussian process X = (X t) t ∈ R , {\displaystyle X=(X_{t})_{t\in \mathbb {R} },} some conditions on its spectrum are sufficient for sample continuity, but fail to be necessary. A necessary and sufficient condition, sometimes called Dudley–Fernique theorem, involves the function σ {\displaystyle \sigma } defined by

-- 02 --

B = K (θ , x ∗ , x ∗) − K (θ , x ∗ , x) K − 1 (θ , x , x ′) K T (θ , x ∗ , x) {\displaystyle B=K(\theta ,x^{*},x^{*})-K(\theta ,x^{*},x)K^{-1}(\theta ,x,x')K^{\mathsf {T}}(\theta ,x^{*},x)}

-- 03 --

A Gaussian process can be used as a prior probability distribution over functions in Bayesian inference. Given any set of N points in the desired domain of the functions, take a multivariate Gaussian whose covariance matrix parameter is the Gram matrix of those N points with some desired kernel, and sample from that Gaussian. For solution of the multi-output prediction problem, Gaussian process regression for vector-valued function was developed. In this method, a 'big' covariance is constructed, which describes the correlations between all the input and output variables taken in N points in the desired domain. This approach was elaborated in detail for the matrix-valued Gaussian processes and generalised to processes with 'heavier tails' like Student-t processes. Inference of continuous values with a Gaussian process prior is known as Gaussian process regression, or kriging; extending Gaussian process regression to multiple target variables is known as cokriging. Gaussian processes are thus useful as a powerful non-linear multivariate interpolation tool. Kriging is also used to extend Gaussian process in the case of mixed integer inputs. Gaussian processes are also commonly used to tackle numerical analysis problems such as numerical integration, solving differential equations, or optimisation in the field of probabilistic numerics. Gaussian processes can also be used in the context of mixture of experts models, for example. The underlying rationale of such a learning framework consists in the assumption that a given mapping cannot be well captured by a single Gaussian process model. Instead, the observation space is divided into subsets, each of which is characterized by a different mapping function; each of these is learned via a different Gaussian process component in the postulated mixture.

-- 04 --

var ■ [X (t)] = E [| X (t) − E ■ [X (t)] | 2] < ∞ for all t ∈ T . {\displaystyle \operatorname {var} [X(t)]={\mathbb {E} }\left[\left|X(t)-\operatorname {E} [X(t)]\right|^{2}\right]<\infty \quad {\text{for all }}t\in T.}

-- 05 --

f (x) = G X g ∼ G P (G X μ g , G X K g G X ′ T) . {\displaystyle f(x)={\mathcal {G}}_{X}g\sim {\mathcal {GP}}({\mathcal {G}}_{X}\mu _{g},{\mathcal {G}}_{X}K_{g}{\mathcal {G}}_{X'}^{\mathsf {T}}).}

-- 06 --

Brownian motion as the integral of Gaussian processes A Wiener process (also known as Brownian motion) is the integral of a white noise generalized Gaussian process. It is not stationary, but it has stationary increments. The Ornstein–Uhlenbeck process is a stationary Gaussian process. The Brownian bridge is (like the Ornstein–Uhlenbeck process) an example of a Gaussian process whose increments are not independent. The fractional Brownian motion is a Gaussian process whose covariance function is a generalisation of that of the Wiener process.

-- 07 --

and maximizing this marginal likelihood towards θ provides the complete specification of the Gaussian process f. One can briefly note at this point that the first term corresponds to a penalty term for a model's failure to fit observed values and the second term to a penalty term that increases proportionally to a model's complexity. Having specified θ, making predictions about unobserved values ■ f (x ∗) {\displaystyle f(x^{*})} ■ at coordinates x* is then only a matter of drawing samples from the predictive distribution p (y ∗ ■ x ∗ , f (x) , x) = N (y ∗ ■ A , B) {\displaystyle p(y^{*}\mid x^{*},f(x),x)=N(y^{*}\mid A,B)} where the posterior mean estimate A is defined as

-- 08 --

σ (h) = 2 E [X t X t] − 2 E [X t X t + h] = 2 ∑ n = 1 ∞ c n 2 sin 2 ■ λ n h 2 . {\displaystyle \sigma (h)={\sqrt {2{\mathbb {E} }[X_{t}X_{t}]-2{\mathbb {E} }[X_{t}X_{t+h}]}}=2{\sqrt {\sum _{n=1}^{\infty }c_{n}^{2}\sin ^{2}{\frac {\lambda _{n}h}{2}}}}.}

-- 09 --

When concerned with a general Gaussian process regression problem (Kriging), it is assumed that for a Gaussian process f {\displaystyle f} observed at coordinates x {\displaystyle x} , the vector of values ■ f (x) {\displaystyle f(x)} ■ is just one sample from a multivariate Gaussian distribution of dimension equal to the number of observed coordinates ■ n {\displaystyle n} ■. Therefore, under the assumption of a zero-mean distribution, ■ f (x ′) ∼ N (0 , K (θ , x , x ′)) {\displaystyle f(x')\sim N(0,K(\theta ,x,x'))} ■, where ■ K (θ , x , x ′) {\displaystyle K(\theta ,x,x')} ■ is the covariance matrix between all possible pairs ■ (x , x ′) {\displaystyle (x,x')} ■ for a given set of hyperparameters θ. As such the log marginal likelihood is:

-- 10 --

whence ∑ n c n (| ξ n | + | η n |) < ∞ {\textstyle \sum _{n}c_{n}(|\xi _{n}|+|\eta _{n}|)<\infty } almost surely, which ensures uniform convergence of the Fourier series almost surely, and sample continuity of X . {\displaystyle X.}

-- 11 --

Rational quadratic: K RQ (x , x ′) = (1 + d 2) − α , α ≥ 0 {\displaystyle K_{\operatorname {RQ} }(x,x')=\left(1+d^{2}\right)^{-\alpha },\quad \alpha \geq 0}

-- 12 --

Stationarity For general stochastic processes strict-sense stationarity implies wide-sense stationarity but not every wide-sense stationary stochastic process is strict-sense stationary. However, for a Gaussian stochastic process the two concepts are equivalent. A Gaussian stochastic process is strict-sense stationary if and only if it is wide-sense stationary.

-- 13 --

Bayesian neural networks are a particular type of Bayesian network that results from treating deep learning and artificial neural network models probabilistically, and assigning a prior distribution to their parameters. Computation in artificial neural networks is usually organized into sequential layers of artificial neurons. The number of neurons in a layer is called the layer width. As layer width grows large, many Bayesian neural networks reduce to a Gaussian process with a closed form compositional kernel. This Gaussian process is called the Neural Network Gaussian Process (NNGP) (not to be confused with the Nearest Neighbor Gaussian Process). It allows predictions from Bayesian neural networks to be more efficiently evaluated, and provides an analytic tool to understand deep learning models.